

RRC Raptor V4H

- Camera

V0.0.2

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VERSION LIST

Description	Version	Date	Author
RRC_Raptor_V4H_Camera is established.	V0.0.1	2024/10/28	KC
Add video test item for chip ID detection	V0.0.2	2025/01/22	KC

HASHTAG

#V4H

#u-boot #kernel

#BSP

#ubuntu

1 Introduction

The documentation shows how to control the video module and display on R-Car V4H2. The video framework is based on V4L2. The schematic diagram of demo is shown as below,

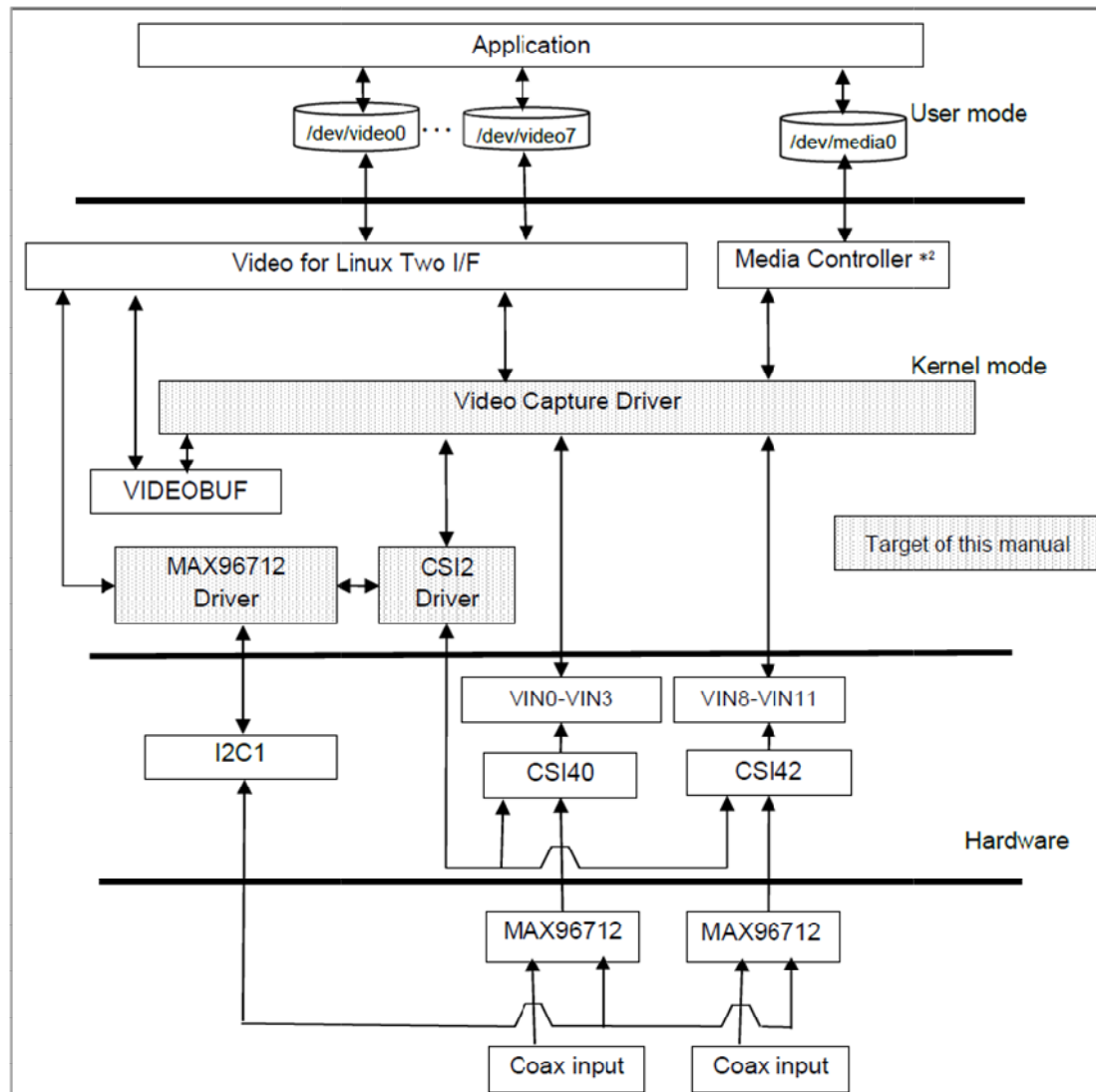


Figure 1-1 Video module configuration of RCar V4H2

1.1 Camera module

The demo is based on LI-AR0231-AP0200-GMSL2-xxxH that is equipped with ON Semiconductor 1/2.7" 2.3MP CMOS digital image sensor AR0231, ISP AP0200 and Maxim serializer MAX9295a. The output image format is YUV422 and Wide Dynamics Range supported.

Item		
Sensor		ON SEMI AR0231
Optical Format		1/2.7"
Maximum Resolution		1920 (H) x 1020 (V)
Pixel Size		3.0 x 3.0 μm
Data Format		YUV422 10-bit
Frame Rate		28.7 fps @ 1920 x 1020
ISP		ON SEMI AP0200
WDR		Supported
Lens	ID	M128B02820WM12R1
	FOV	148°(D) / 122°(H) / 74°(V)
	EFL (mm)	2.8
Serializer		Maxim MAX9295A/B
Power Supply Range		9 ~ 19 VDC
Connector		FAKRA Z TYPE
IP Rating		IP67
Power Consumption		Approx. 116 mA @ 12 VDC
Operating Temp		-40°C ~ +85°C
Storage Temp		-40°C ~ +105°C
Mount		M12

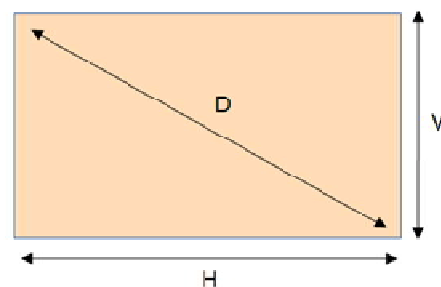


Table 1-1 Feature of AR0231

➤ **Camera and GMSL2 Fakra cable**



Figure 1-2 Camera and GMSL2 cable

1.2 V4L2 API

Function name	Description	Remark
ioctl(VIDIOC_QUERYCAP)	Query device capabilities.	#1
ioctl(VIDIOC_REQBUFS)	Buffer demand and transmission mode select.	#1
ioctl(VIDIOC_G_FMT)	Get a format.	#1
ioctl(VIDIOC_S_FMT)	Set scaling and the data format	#1
ioctl(VIDIOC_TRY_FMT)	Try a format.	#1
ioctl(VIDIOC_QUERYCTRL)	Enumerate controls and menu control items.	#3
ioctl(VIDIOC_G_CTRL)	Get the value of a decoder control.	#3
ioctl(VIDIOC_S_CTRL)	Set the value of a decoder control.	#3
ioctl(VIDIOC_CROPCAP)	Information of the video cropping and scaling abilities.	#1
ioctl(VIDIOC_G_CROP)	Get the current cropping rectangle.	#1
ioctl(VIDIOC_S_CROP)	Set the current cropping rectangle.	#1
ioctl(VIDIOC_QUERYBUF)	Query the status of a buffer.	#2
ioctl(VIDIOC_DQBUF)	Dequeue an empty buffer.	#2
ioctl(VIDIOC_QBUF)	Enqueue a filled buffer.	#2
ioctl(VIDIOC_STREAMON)	Start streaming I/O.	#2
ioctl(VIDIOC_STREAMOFF)	Stop streaming I/O.	#2

Notes: #1 This module provides these ioctls.

#2 VIDEObUF provides these ioctls.

#3 These ioctls is provided by subdevice (/dev/v4l-subdev1 or /dev/v4l-subdev2).

Table 1-2 V4L2 API list

1.3 Terminology

The common terminologies are as described bellows,

Title	Location
V4L2	Video for linux 2
VIN	Video input module
MIPI	Module industry processor interface
CSI-2	Camera serial interface 2
CSI40	4 lanes of CSI-2 LINK 0
CSI41	4 lanes of CSI-2 LINK 1
CSI42	4 lanes of CSI-2 LINK 2
CSI43	4 lanes of CSI-2 LINK 3
SerDes	Serializer and Deserializer

Table 1-3 Terminology

2 Video test

The chapter shows how to check the mounting status of devices through i2c-tool.

➤ SerDes MAX96712

\$ i2cdetect -r -y 1

```
root@v4x:/sample/dms_ref_app# i2cdetect -r -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  --  --  --  --  --  28  --  2a  --  UU  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40: 40  --  --  --  --  --  --  --  --  UU  --  UU  --  --  --
50: UU  UU  UU  UU  UU  --  --  --  --  --  --  --  5d  --  --
60: UU  --  --  --  --  --  --  --  --  --  --  --  --  --  --
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
```

MAX96712 chip ID

\$ i2ctransfer -f -y 1 w2@0x49 0x00 0x0D r1

\$ i2ctransfer -f -y 1 w2@0x4B 0x00 0x0D r1

[OUTPUT] 0xA0

➤ MAX20087

\$ i2cdetect -r -y 1

```
root@v4x:/sample/dms_ref_app# i2cdetect -r -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  --  --  --  --  --  28  --  2a  --  UU  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40: 40  --  --  --  --  --  --  --  --  UU  --  UU  --  --  --
50: UU  UU  UU  UU  UU  --  --  --  --  --  --  --  5d  --  --
60: UU  --  --  --  --  --  --  --  --  --  --  --  --  --  --
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
```

MAX20087 chip ID

\$ i2ctransfer -f -y 1 w1@0x28 0x02 r1

\$ i2ctransfer -f -y 1 w1@0x2A 0x02 r1

[OUTPUT] 0x21

➤ VIN

Show all media device via v4l2 framework.

```
$ v4l2-ctl --list-devices
```

```
R_Car_VIN (platform:e6ef0000.video):  
    /dev/video0  
    /dev/media0  
  
R_Car_VIN (platform:e6ef1000.video):  
    /dev/video1  
  
R_Car_VIN (platform:e6ef2000.video):  
    /dev/video2  
  
R_Car_VIN (platform:e6ef3000.video):  
    /dev/video3  
  
R_Car_VIN (platform:e6ef8000.video):  
    /dev/video4  
  
R_Car_VIN (platform:e6ef9000.video):  
    /dev/video5  
  
R_Car_VIN (platform:e6efa000.video):  
    /dev/video6  
  
R_Car_VIN (platform:e6efb000.video):  
    /dev/video7
```

2.1 Media device topology

v4l2-media framework analyze each device into an abstract module and link each other. The data stream between each device will be connected. The topology is shown as described bellows,

a) Topology of /dev/media0

[Raptor] \$ media-ctl /dev/media0 --print-dot > media0.dot

b) Download the essential package

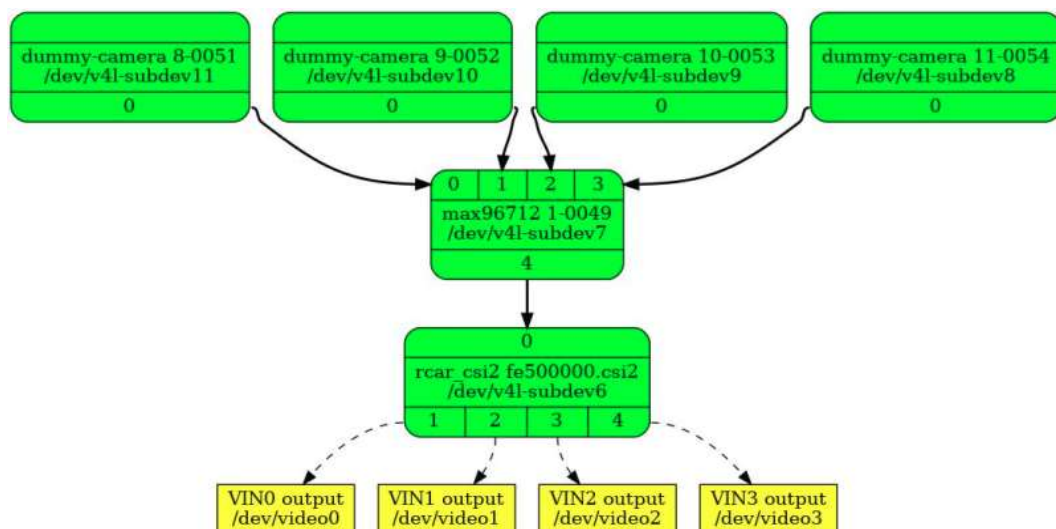
[HOST] \$ sudo apt-get install -y graphviz

c) Show the topology image

Convert the dot file to a visualized figure by python as following,

```
from graphviz import Source

if __name__ == "__main__":
    path = 'media0.dot'
    s = Source.from_file(path)
    s.render(filename='media0',
            directory=".",
            format='jpg',
            view=True)
```



2.2 Front camera system

Reference : Renesas\rcar-xos\v3.18.0\docs\sw\frontcam_ref_app\R-CarV4H2_FrontCam_ref_app_User_Manual.pdf

➤ Configuration

```
$ vi config/frontcam_customize.config
```

```
----Configuration----
VIN_Enable           = 1
VIN_Device           = 0      # 0 ~ 7 - (/dev/videoX)
VIN_Capture_Format   = 3      # 0:"YUYV", 1:UYVY, 2:RGB24, 3:Y10
Frame_Width          = 1920
Frame_Height         = 1020
VOUT_Enable          = 1
```

➤ Run app

```
$ frontcam_ref_app_v4h2
```

➤ Run app with CDNN load

```
$ frontcam_with_cdnn_ref_app_v4h2
```

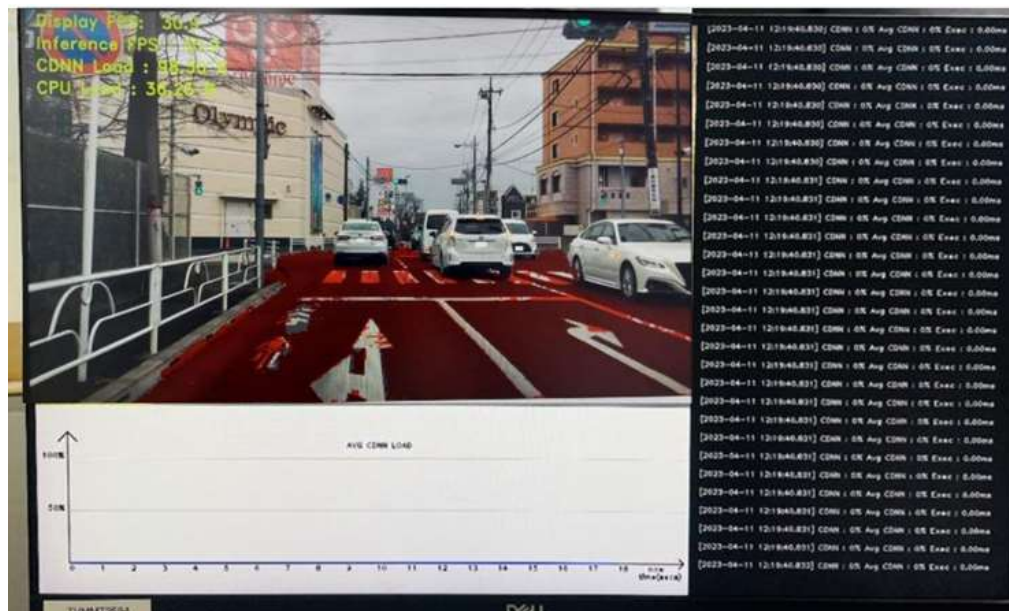


Figure 2-1 Frontcam system with CDNN

2.3 Driver monitoring syste

Reference : Renesas\rcar-xos\v3.18.0\docs\sw\dms_ref_app\R-
CarV4H2_DMS_ref_app_User_Manual.pdf

➤ Configuration

\$ vi config/dms_customize_v4h.config

```

----Configuration----
VIN_Enable           = 1
VIN_Device           = 0      # 0: VIN0~VIN3, 4: VIN8~VIN11
VIN_Capture_Format   = 3      # 0:"YUYV", 1:UYVY, 2:RGB24, 3:Y10
Frame_Width          = 1920
Frame_Height         = 1020
VOUT_Enable          = 1
  
```

➤ Run app

\$ dms_ref_app_v4h2

➤ Run app with CDNN load

\$ dms_with_cdn_ref_app_v4h2

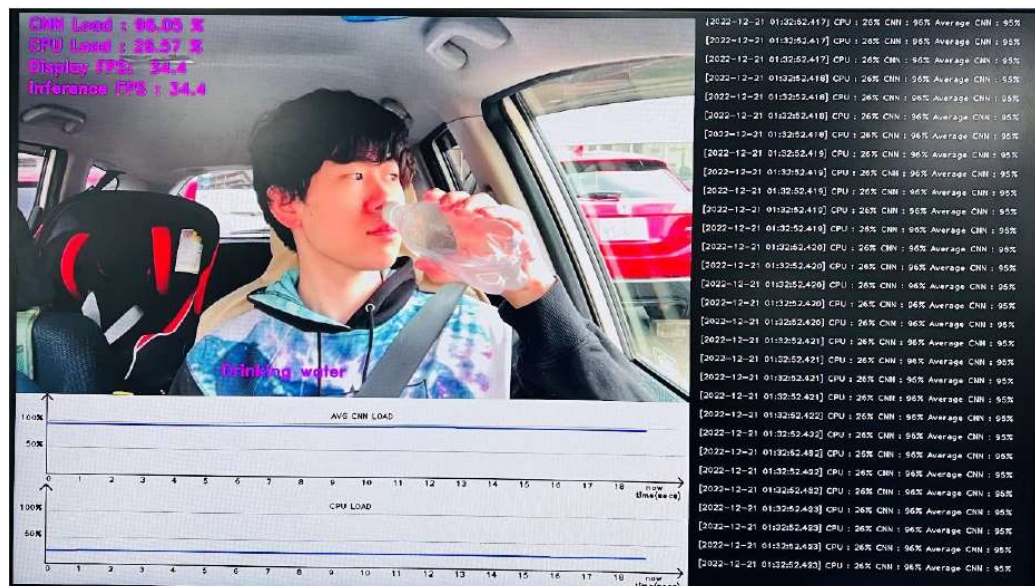


Figure 2-2 Driver monitoring system with CDNN

2.4 Surrounding view system

Reference : [Renesas\rcar-xos\v3.18.0\docs\sw\surroundview_ref_app\R-CarV4H2_SurroundView_ref_app_User_Manual.pdf](#)

➤ Configuration

```
$ vi config/surroundview_customize.config
```

```
----Configuration----
VIN_Enable           = 1
VIN_Device           = 0      # 0: loop all /dev/video0 ~ 7
VIN_Capture_Format   = 3      # 0:"YUYV", 1:UYVY, 2:RGB24, 3:Y10
Frame_Width          = 1920
Frame_Height         = 1020
VOUT_Enable          = 1
```

➤ Run app

```
$ surroundview_ref_app_v4h2
```

➤ Run app with CDNN load

```
$ surroundview_with_cdn_ref_app_v4h2
```

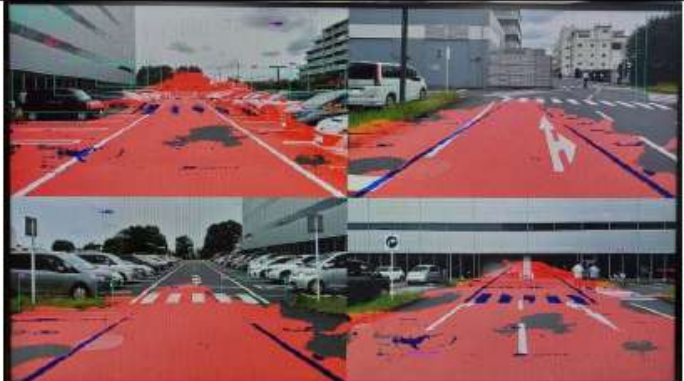
No.	Enabled Feature	Output example
1	Semantic Segmentation	
2	Object Detection	
3	Semantic segmentation + Object detection	

Figure 2-3 Surroundview with CDNN